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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CHM 111 — Past Questions & Materials
by Renance

Course: CHM 111
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 5
Levels: 100

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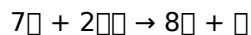
Paper 1 of 5 — CHM 111 TEST

Level 100 • First Semester • PastQuestion

CHM 111 (GENERAL INORGANIC CHEMISTRY) TEST

1. In the nuclear reaction represent below, what does X represent?

${}^4_2\text{He} + {}^{12}_6\text{C} \rightarrow {}^{16}_8\text{O} + \text{X}$



Ans: ${}^{11}_1\text{H} = {}^1_1\text{H}$ Proton

2. What is another name for Beta particle?

Ans: Electron emission

3. Why is cryolite added to aluminum ore in extraction process?

Ans: To reduce its melting point

4. Many metals are found in the earth's crust ore. What is an ore?

Ans: A mineral deposit concentrated enough to allow economical recovery of a desired metal is known as ore (Chemistry by RAYMOND CHANG).

According to Wikipedia, An ore is a type of rock that contains sufficient minerals with important elements including metals that can be economically extracted from the rock.

5. State the modern periodic law

Ans: It states that the chemical and physical properties of an element are periodic function of their atomic number.

6. Why do transition metals possess variable oxidation states?

Ans: Transition metals possess variable oxidation states because of their ionization energy.

(The answer is not well detailed read RAYMOND CHANG: Chemistry pg 956-967.)

Complete the table

S/N Atomic Number Chemical Name Group Period

Number Symbol

7. 11 Na Sodium 1 3

8. 14 Si Silicon 4 3

9. 20 Ca Calcium 2 4

10. Name six elements that show anomalous behavior

Ans: Li – Lithium, Mg – Magnesium, Be – Beryllium, Al – Aluminum, B – Boron, Si – Silicon

11. Write the formulas for the following

I. Potassium hydroxide

Ans: KOH

II. Boron trifluoride

Ans: BF₃

12. Name the various four quantum numbers which completely specify the electrons of the

atom.

Ans: i.) Principal Quantum number

ii.) Angular / Azimuthal Quantum number

iii.) Magnetic Quantum number

iv.) Spin Quantum number

13. Name the Principal / rules established for order of building electronic structures

Ans: i.) Pauli Exclusion Principle

ii.) Hund's Rule

iii.) Aufbau Principle

14. Give the formula of a compound formed when aluminum combine with oxygen

Ans: Al_2O_3

15. What is the trend for atomic radii down a group and across the period in the periodic table?

Ans: Atomic radii increase down the group and decreases from left to right across the period in the periodic table.

16. Write down the number of 3d electrons in each of the following ions

Cr^{2+} , Cr^{3+} , Fe^{2+} Fe^{3+}

Ans:

i.) Cr^{2+} : $[Ar] 4s^0 3d^4$

ii.) Cr^{3+} : $[Ar] 4s^0 3d^3$

iii.) Fe^{2+} : $[Ar] 4s^0 3d^6$

iv.) Fe^{3+} : $[Ar] 4s^0 3d^5$

Paper 2 of 5 — CHM 111 2016/2017

Level 100 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN, NIGERIA

DEPARTMENT OF CHEMISTRY

FIRST SEMESTER EXAMINATION — 2016/2017 SESSION

COURSE CODE/TITLE: CHM 111; GENERAL INORGANIC CHEMISTRY TIME: 14HOURS

INSTRUCTION: ANSWER ALL QUESTIONS

OPTION C

1, An element Q has 180 neutrons and 70 electrons, what is its atomic number? 1

2. The elements in Groups 1 and 2 are known respectively as..... ?

3 in the missing atomic numbers,

Use the Table below to answer questions 3-5. Complete the table below, fill in chemical symbols and names.

Atomic Chemical Name Group Period

Number Symbol

3: 11 Sodium

4. 14 Si :

5, 20 | Calcium |

6. What is the maximum number of electrons that can be placed in a single 3d orbital?"

7. The simplest formula for a compound containing Mn⁴⁺ and O²⁻ is ?

8, Name six elements that show anomalous behaviour

9, What are the possible numbers of electrons in a stable f- orbital?

10. The atomic number of an element is 11 and its mass number is 23. The number of protons, electrons and

neutrons present in the neutral atom are..... :

11. Name the various four quantum numbers which completely specify the electron of an atom.

12. Name the principles/ rules established for order of building electronic structures...

13. Give the sequence of filling energy levels in atoms up to the 7th main energy levels.

14. The sum of the number of protons and neutrons in the nucleus of an atom is called the...

15 .The nucleus is composed of ... and ... particles

16. List the three different modes of beta decay

17. Why is cryolite added to the aluminium ore in the extraction process?

18, Many metals are found in the Earth's crust as ores. What is an ore?

19. Why are the alkali metals stored under oil?

20. What type of bond will form between two non-metals that differ slightly in electronegativity?

21. The energy required to separate the ions in one mole of an ionic solid is

22. Who initially proposed arrangement of elements by atomic numbers, which we still use today?
23. State the Modern Periodic Law.
24. Why do transition metals possess variable oxidation states?
25. What are the three principal steps in metal extraction?
26. What is the hybrid orbital formed when one S and two P orbitals mix up? sp^2
27. Calculate the number of atoms in a molecule of ammonium tetraoxosulphate(vi), $(NH_4)_2SO_4$
28. What is inert pair effect? ns^2
29. Give the formula of a compound formed when aluminum combines with oxygen.
30. What is the trend for atomic radii down a group and across the period in the periodic table?
31. Electrons that are not used in bonding are known as lone pair electrons
32. Why is the bond angle of ammonia molecule less than that of a methane molecule even though they both have four pairs of electron around their central atoms?
33. Why is the 2nd ionization energy of K so much higher than its 1st ionization energy?
34. State the products for the following chemical reactions
- (i) $Si + O_2 \rightarrow$
- (ii) $B_2O_3 + NaOH \rightarrow$
35. Electronic configuration of a transition element, Y in +2 oxidation state is $[Ar]3d^5 4s^0$ | What is its atomic number?
36. Write down the electronic configuration of each of the following ions: Zn^{2+} , Cr^{3+} , Mn^{2+} and Fe^{2+}
37. What is the maximum oxidation state shown by manganese?
38. List two general physical characteristics of transition metals
39. Find X in given reaction
- $2K + 2H_2O \rightarrow 2KOH + X$
40. Name the element in group IIA that is a non-metal and the only member whose compounds are all covalent
41. The most important way carbon differs from other elements of group IVA is.....
42. Write the formula of the gas liberated when group IIA reacts with cold water?
43. Name the element in group IIA that tarnishes and forms a film of protective oxide when exposed to air
44. Name the two elements in group IVA that are metalloids.
45. Name the element in group IIA that is prepared by carbon reduction of its oxide from sea water.
46. What is the method used to obtain aluminum from bauxite?
47. Lead reacts with air to form a thin coating of &/or....which protects the metal from further reaction
48. Which of the elements in the first row of the transition metal series has properties that make it useful for electrical wiring and plumbing?

49. In the nuclear reaction represented below, what does X represent?

${}^y_4\text{He} \longrightarrow$ *

50. What is another name for the beta particle?

51. What is responsible for the insolubility of the salts of the alkaline earth metals?

52. The most likely mode of decay for a neutron-rich nucleus is one that converts a neutron into

x

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END OF PAPER

Paper 3 of 5 — CHM 111 2015/2016

Level 100 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN, DELTA STATE
COLLEGE OF SCIENCE
DEPARTMENT OF CHEMISTRY

COURSE TITLE: GENERAL INORGANIC CHEMISTRY CODE: CHM 111
FIRST SEMESTER EXAMINATION (2015/2016 SESSION) TIME: 1½ HOURS
INSTRUCTION: ANSWER ALL QUESTIONS OPTION C

1. An alkaline earth element (Group IIA) is radioactive. It undergoes decay by emitting three α -particles in succession. In what periodic table group is the resulting element found? 2. Name the two impurities found in the ore of bauxite? 3. What happens when i) sodium ii) boron are exposed to air? 4. Give three examples of Group (IVA) elements. 5. Why are the alkali metals not found in the elemental state? 6. Give the chemical formula of cryolite. 7. List the general procedures for extraction of metals. 8. Why is the polarizing power of group IIA much greater than that of group IA
9. Give two reasons why it is difficult to prepare elemental boron in a high state of purity
10. List the pairs of elements in groups IA-IVA that show diagonal relationships.
11. Boron reacts readily with steam to form compound A and liberate gas B. Write a balanced chemical equation for this reaction.
12. The mass of one mole of electron in milligrams is --- (assume mass of one electron to be 9.1×10^{-31} kg)
13. Copper prefers this electronic configuration $[\text{Ar}] 4s^1 3d^{10}$ why
14. The colour of a Mn compound in +7 oxidation state is
15. The most important oxidation states of iron are +2
16. Transition metals form metal complexes with-
17. Give the last five elements of the first transition series
18. The half-life of ^{11}C is 20.3 min. How long will it take for 95.0% of a sample to decay?
19. How is positron emitted in a nuclear reaction

This diagram shows the electrolytic extraction of aluminum from bauxite. Use the diagram to answer

questions 20- 22.

20. Identify A and E
21. What are the constituents of B
22. What is the physical state of D
23. What method is used for the extraction of magnesium from seawater? electrolysis
24. Give the formula of Borax $\text{Na}_2\text{B}_4\text{O}_7$
25. Which of the elements in group IA forms a stable nitride? Name and give the formula of the nitride Li_3N , Li
26. Name one element that appears in group IA that is not an alkali metal.
27. Why is potassium more reactive than lithium? it is less electronegative than potassium
28. What is the cause of the anomalous behavior of lithium?
29. Why is aluminum metal not extracted by carbon reduction of its oxide?
30. In a reaction with a non-metal, each alkali metal atomA..... its outer electron and becomes anB..... with a ...C.....charge.
31. Indicate which ion in the following pairs is larger in size: (a) N^{3-} or F^- (b) Mg^{2+} or Ca^{2+} (c) Fe^{2+} or Fe^{3+}
32. Define inert pair effect
33. Write the theoretical expression for the determination of electronegativity
34. Write the three degenerate orbitals of the P- subshell
35. How many unpaired electrons are in one atom of iron, $^{56}_{26}\text{Fe}$

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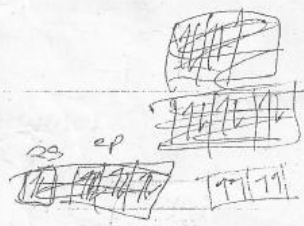
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36. What is the atomic number of an element with the following electronic configuration: $1s^2 2s^2 2p^6 3s^2 3p^4 4s^2 3d^{10}$
37. How are atomic radius and effective nuclear charge related to electronegativity
38. Ammonia has a bond angle of 107.3° while methane has a bond angle of 109.5° . Why the difference in their structural geometries. 39. Write the excited state configuration of carbon atom. $[\frac{12}{6}\text{C}]$. 40. In which blocks of the periodic table do these elements belong to: Na, Sc and Cl. 41. Why are +4 ions in group IVA or XIV unknown
42. In chromium the $3d^4 4s^1$ electron configuration is less stable than the $3d^5 4s^1$; why?
43. How many unpaired electrons are in one atom of iron, $^{56}_{26}\text{Fe}$
44. How many neutrons and protons are contained in the atom of gold, $^{197}_{79}\text{Au}$
45. Write the values of the magnetic quantum number, M_l for the value of subsidiary quantum number, $\ell = 3$.
46. Write the formula of the compound formed between sodium and a group VII element, X.
47. What is the standard unit of radioactivity? 48. What is Nuclear Fission?
49. What is the atomic number of a β^- particle? 50. Give two ways of determining a stable nucleus.
51. Which transition metal has the highest number of oxidation states?
52. When is a transition metal said to be paramagnetic?
53. A nuclide of the element, rutherfordium, $^{257}_{104}\text{Rf}$, is formed by the nuclear reaction of californium-249, $^{249}_{98}\text{Cf}$ and carbon-12, $^{12}_6\text{C}$ with the emission of four neutrons. This new nuclide rapidly decays by emitting an α -particle. Write the equation for each of these nuclear reactions.
54. The half-life of $^{19}_8\text{O}$ is 29 s. What fraction of the isotope originally present would be left after 10.0 s?
55. Write the nuclear equation for the following process: Two deuterium ions undergoing fusion to give ^3_2He and a neutron
56. How many neutrons and protons are contained in the atom of gold, $^{197}_{79}\text{Au}$
57. List the four quantum numbers
58. Write the number of electrons and charge of the ion, $^{53}_{25}\text{Mn}^{2+}$
59. To which group of the periodic table does the element with the following electronic configuration belong to: $\text{Ne} 3s^2 3p^2$?
60. State the Hund's rule
61. What are the shape and bond angle of beryllium hydride, BeH_2
62. Define hybridization
63. List the hybrid orbitals of carbon atom when it undergoes hybridization
64. What is the maximum number of electrons in the principal quantum number, $n = 4$
65. What is periodicity
66. What is the total number of values of subsidiary quantum number when the principal quantum number is 4?
67. Arrange the first ionization energy, I_1 , second ionization, I_2 , third ionization energy, I_3 in order of decreasing energy.
68. $\text{X(g)} + \text{e}^- \rightarrow \text{X}^-(\text{g}) + \text{energy}$. What name is given to the energy?
69. What is the representative name for the group IIA elements
70. Which of the following elements is not a d-block element: Sc, Ni, V, Ni, Cu and Zn

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Paper 4 of 5 — CHM 111 2014/2015

Level 100 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLUUM RESOURCES, EFFURUN, NIGERIA

DEPARTMENT OF CHEMISTRY

FIRST SEMESTER EXAMINATION, 2014/2015 SESSION. JUNE, 2015

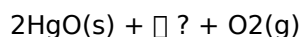
COURSE CODE/TITLE: CHM 111; General Inorganic Chemistry – 3 Credit Units

INSTRUCTION: Answer All Questions. TIME: 1Hour 30Minutes

OPTION A

1. What is the difference in terms of thermal stability between lithium trioxocarbonate (iv) and potassium trioxocarbonate (iv).
2. What is the product of the reaction below
 $3M + N_2$ Where M is an alkaline earth metal
3. Why is aluminum resistant to attack by dilute hydrochloric acid .
4. The electronic configuration of indium, atomic number, 49 is?
5. Why are the group 3A elements said to be Lewis acids?
6. Thallium oxide has no acidic character unlike the rest of the 3A oxides. Why?
7. List four elements whose hydroxides are amphoteric in character?
8. Why are +4 ions unknown in group IV A elements?
9. Why is catenation possible in carbon?
10. State for elements that exhibit catenation apart from carbon.
11. Why does zinc lack many typical properties of Transition Metals
12. Which is the Lewis base in the chemical equation represented below? $D + A \rightleftharpoons D:A$
- 13 What is the colour of the ion, Cu^{2+}
13. What is responsible for magnetism?
15. State one serious disadvantage of iron as a structural material
16. What is the percentage of water in $Na_2CO_3 \cdot 5H_2O$
17. What is the difference in the number of electrons of Ca and Ca^{+}
18. Why is iron paramagnetic and zinc is not
19. In the extraction of aluminum, 40 tonnes of the ore, bauxite was fed into the extraction system, instead of the normal 3.6 tonnes. What is the mass error in Kilogram?
20. Write the chemical formula for the mineral, Trona and state the oxidation number of oxygen atoms in a mole of the mineral.
21. What is hybridization?
22. How many unpaired electron(s) has Na atom

23. Which two criteria did Mendeleev use to classify elements in the periodic table?
24. Give the general trend in electronegativity
25. What are the products resulting from the action of ionization energy on a gaseous atom?
26. The hybridization of orbitals on the central atom in a molecule is sp . What is the electron domain geometry around the central atom?
27. Which of the quantum numbers is used to characterize the shape of orbitals?
28. What is responsible for sharp decrease in ionization energy from Boron to Aluminum?
29. How many electrons exist in the penultimate shell of Zinc
30. Boron combines with metals to form while Aluminum forms?
31. Use chemical equations to explain the amphoteric nature of Beryllium Oxide
32. Give a simple method you can use to identify a Barium metal and Magnesium metal
33. Mention the two metals that can be found in a Beryl Ore
34. What is inert pair effect?
35. Using chemical equations, show the reaction of magnesium with nitrogen
36. A group 2 metal M, dissolves in NH_3 and from the solution, compound A can be isolated with the liberation of gas C. Metal M gives a crimson flame test. Suggest the identities of M, A and C
37. Mention one use of Barium Sulphate.
38. Explain the thermal stability of hydrides of the group 15 elements
39. Write the formulas of metaplumbic acid and carbonic acid
40. Elements with the mass number but different neutron and proton numbers are called
41. What is a ligand?
42. Give an example of a purple coloured manganese compound and the oxidation state of Mn in the compound
- 43.
44. An oxide of transition metal used as a catalyst in the contact process is?
45. Why is the electronegativity of Al and Ga almost the same
46. The number of orbitals in a f-sub-shell is
47. The atomic number of an element is 17 and its mass number is 35. The number of protons, electrons and neutrons present in the atom are.....?
48. The maximum number of electrons that can be accommodated in the nth level is?
49. State the two principal methods involved in concentration of metal ores
50. Complete the chemical equation below for the thermal decomposition of mercuric oxide



51.

52. What is the other product formed when potassium-40 decays to produce calcium-40?

53. What is the charge on Gamma ray?

54. The order of penetrating power of particles from lowest to highest is ?

55. Which of the sub atomic particles is not a nucleon?

56. Define nuclear fusion?

57. The nuclear reaction used to produce energy for nuclear bomb is ?

58. What type of rays can cause genetic changes?

59. An alpha particle is actually a helium _____ ?

60. _____ travels at a speed of 3×10^8 m/s

61. How does the ionic radius of K^+ compare to the atomic radius of K

62. What is the product resulting from the action of electron affinity and an electron on a gaseous atom?

63. Arrange the following in order of increasing radius, O^{2-} , Ne, F^-

64. How would the energy necessary to convert a gaseous Al atom to gaseous a gaseous Al^{3+} ion be found?

65. Why is the electron configuration of chromium $1\text{S}^2 2\text{S}^2 2\text{P}^6 3\text{S}^2 3\text{P}^6 4\text{S}^1 3\text{D}^5$ an exception to the Aufbau principle?

66. Write the excited state configuration of carbon

67. How many electrons are required for the stability of an f-orbital

68. What is the values of magnetic quantum number,

69. State the rules/principles used for electronic configuration of an atom of an al element.

70. List two metallurgical industries in Nigeria

Paper 5 of 5 — CHM 111 2010/2011

Level 100 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN, NIGERIA
DEPARTMENT OF CHEMISTRY
FIRST SEMESTER EXAMINATION, 2010/2011 ACADEMIC SESSION JULY, 2011
CHM111: GENERAL INORGANIC CHEMISTRY
INSTRUCTIONS: ANSWER ALL QUESTIONS. TIME: 1 HOUR 30 MINUTES
OPTION A

- A necessary prerequisite to the construction of the periodic table was
A. the discovery of phosphorus B. the discovery of individual elements C. the discovery of atomic number D. the discovery of gold E. None of the above
- The forerunners to Mendeleev's scheme of classification of the periodic chart are
A. Johann Dobereiner and John Newlands B. Dmitri Ivanovich and Ivanovich Mendeleev C. John Newland and Ivanovich Mendeleev D. Hennig Brand and John Newland E. None of the above
- Among the fundamental atomic properties, the first ionization energy is the best indicator of whether an element behaves
A. as a metalloid B. as a nonmetal C. as a metal D. as a gas E. None of the above
- A radioactive isotope has a half-life of 30 minutes. If we have 6.00 milligram sample of this isotope at 12.00pm. How much will be left at 2.00 pm?
A. 3.00 mg B. 1.50 mg C. 0.375 mg D. 0.750 mg E. None of the above
- The law of triads, one of the earliest attempts to trace interdependence between the properties of elements and their atomic masses was put forward by
A. Hennig Brand B. Beguyer de Chancourtois C. John Dobereiner D. Lothar Meyer E. None of the above

Element	Atomic Mass
Cl	35.5
Br	80
I	127

From the table above, the supposed atomic mass of Br according to the law of triad is
A. 103.3 B. 83 C. 81 D. 82 E. None of the above

- The atom formed by the beta decay of carbon-14 is
A. oxygen-18 B. beryllium-10 C. boron-14 D. nitrogen-14 E. None of the above
- The half-life of francium-212 is 19 minutes. How many minutes will it take for 4.75 mg of this isotope to decay to 0.125 mg?
A. 4.75 minutes B. 9.5 minutes C. 38 minutes D. 57 minutes E. None of the above
- Newland's ideas on the classification of the periodic table were criticized because
A. it assumed that all elements had been discovered B. it placed all the elements appropriately C. Aluminium and Potassium were grouped together D. Silicon and Fluorine were grouped together E. None of the above
- Mendeleev's periodic table was more acceptable than those proposed earlier because
A. it took care of all the elements discovered so far B. it considered the vertical grouping of elements adequately C. it considered the horizontal grouping of elements sufficiently D. it made provision for elements yet to be discovered E. None of the above
- The modern form of the periodic law states that
A. the properties of elements are a periodic function of their mass number B. the properties of elements are a periodic function of their melting point C. the properties of elements are a periodic function of their atomic radius D. the properties of elements are a periodic function of their atomic volume E. None of the above
- Fluorine atoms tend to when they form chemical compounds with metals.
A. lose electrons B. gain electron C. neither lose nor gain electrons D. fluorine atoms do not form compounds with other atoms E. None of the above
- Which of the following electronic configurations represents the electronic configuration of magnesium cation?
A. $1s^2 2s^2 2p^4 3s^2 3p^2$ B. $1s^2 2s^2 2p^4 3s^2 3p^2$ C. $1s^2 2s^2 2p^4$ D. $1s^2 2s^2 2p^6$
- What do the following have in common? ^{20}Ne , ^{19}F , $^{24}\text{Mg}^{2+}$

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- They are isotopes of each other B. They are isomers of each other. C. They are electronic with each other D. They are different elements, so they have nothing in common E. None of the above
15. Which of the following has the largest diameter?
A. F B. Cl C. Br D. I E. None of the above
16. Which of the following has the greatest electronegativity?
A. Si B. P C. N D. O E. None of the above
17. Which of the following has the smallest diameter?
A. O^{2-} B. Na C. F D. Al^{3+}
18. The following are d-block elements except
A. Zn B. Se C. Cu D. Pt E. None of the above
19. p -orbitals are A. dumb bell shaped B. spherical in shape C. double dumb bell in shape D. pyramidal in shape E. None of the above
20. ns^2p^3 configuration is the configuration of
A. group 3 elements B. group 2 elements C. group seven elements D. group 5 elements E. none of the above
21. Which of these electronic configurations of a transition metal is the most stable?
 $ns^2p^6d^1$ B. $ns^2p^6d^3$ C. $ns^2p^6d^4$ D. $ns^2p^6d^5$ E. None of the above
22. Which of the following atoms will form a diatomic molecule?
A. B B. Na C. F D. Ar E. None of the above
23. The element with the greatest tendency to form covalent compound with halogen is
A. Ca B. Al C. Na D. C E. None of the above
24. The number of valence electrons in an element with atomic number 16 is
A. 4 B. 6 C. 5 D. 2 E. None of the above
25. Which of the following types of bonding does not involve the formation of a new substance?
A. Coordinate B. Covalent C. Metallic D. Electrovalent E. None of the above
26. One of the following oxides does not exist:
A. Li_2O B. Rb_2O C. NaO_2 D. Na_2O E. None of the above
27. Energy + $X(g) \rightarrow X^+(g) + e^-$ I_1
Energy + $X^+(g) \rightarrow X^{2+}(g) + e^-$ I_2
Energy + $X^{2+}(g) \rightarrow X^{3+}(g) + e^-$ I_3
X = element, I_1 , I_2 , I_3 are 1^{st} , 2^{nd} & 3^{rd} ionization energies.
From the above
 $I_1 = I_2$ B. $I_1 = I_2 + I_3$ C. $I_1 < I_2$ D. $I_1 > I_2$ E. None of the above
28. The variation of ionization energy in the following period 2 elements: Be, B, N, O can be represented thus:
A. $Be < B < N < O$ B. $Be < B < O < N$ C. $Be > B < O > N$ D. $Be > B < O < N$ E. None of the above
29. Which of the following elements has the highest number of spins?
A. Nitrogen B. Oxygen C. Fluorine D. Neon E. None of the above
30. The order of increasing reducing powers of the alkali metals is:
A. $Cs < Rb < K < Li$ B. $Li < K < Rb < Cs$ C. $Li < Cs < Rb < K$ D. $K < Li < Cs < Rb$ E. None of the above
31. CH_4 is inert but SiH_4 is easily hydrolysed by water. Give a reason:
A. Si has a larger atomic radius; B. Si has d -orbital to accept electron pairs; C. Carbon is more electronegative; D. None of the above
32. Which of the following element does not belong to the family indicated?
A. Rubidium (Rb) - alkali metal B. Selenium (Se) - halogen C. Barium (Ba) - alkaline earth metal D. Ruthenium (Ru) - transition metal E. None of the above
33. Which of the following groups of elements have been placed in order of increasing metallic character?
i. Na Rb Cs ii. P S Cl iii. Al Si Cu iv. O S Se
A. (i) and (iv) B. None of them C. (i) and (ii) D. (i) only E. None of the above
34. In which of the following are the different species placed in the correct order of decreasing size?
A. $Na^+ > Na$ B. $I^- > I$ C. $O > O^{2-}$ D. $Fe^{3+} > Fe$
35. Which of the following isoelectronic ions would you expect to have the smallest radius?

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36. Which of the following contains an error?
 A. Alkali metal - Cs, Li, B. The coinage metals are Au, Ag and Cu C. Transition elements, Sn and Pb can exhibit variable valencies D. Liquid halogens - I_2 , Br_2 E. None of the above
37. The first ionization energy of the element M can be represented by:
 A. $M(g) \rightarrow M^+(g) + 2e^-$ B. $M(g) \rightarrow M^+(g) + e^-$ C. $M^+(g) + e^- \rightarrow M(g)$ D. $M(g) \rightarrow M^+(g) + e^-$
38. Which of the following values represent the electron affinity of Chlorine?
 A. 348 kJ/mole B. -348 kJ/mole C. $+348 \text{ kJ/mole}$ D. $\pm 348 \text{ kJ/mole}$ E. None of the above
39. If $X = I.E. + E.A.$ Where I.E. = ionization energy and E.A. = electron affinity, X is
 A. electronegativity B. electropositivity C. atomic volume D. atomic radius E. None of the above
40. Shapes of some molecules depend on the
 A. electronic configuration of the constituent elements B. bonding type C. electron pairs D. orientation of orbitals in space E. None of the above
41. The shape of phosphorus pentachloride (PCl_5) is
 A. linear B. tetrahedral C. pyramidal D. trigonal bipyramid E. None of the above
42. The bond angle in ammonia molecule is less than that of a normal tetrahedral molecule because
 A. it has four bonding pairs of electrons B. it has two bonding pairs and two lone pairs C. it has one bonding pair and three lone pairs D. it has three bonding pairs and one lone pair E. None of the above
43. Water molecule is V shaped (distorted tetrahedron) because of
 A. the effect of one lone pair of electrons B. the effect of two lone pairs of electrons C. the effect of three lone pairs of electrons D. the effect of four lone pairs of electrons E. None of the above
44. sp^3 hybrid molecules are formed by the mixing of
 A. one s-orbital and three p-orbitals B. four p-orbitals C. one p-orbital and three s-orbitals D. two p-orbitals and two s-orbitals E. None of the above
45. The equivalent of single bonds of the electron pair repulsion theory in the orbital theory is
 A. σ bond B. π bond C. d-bond D. x-bond E. None of the above
46. An example of sp hybrid molecule is
 A. H_2 B. F_2 C. HF D. NH_3 E. None of the above
47. The bond angle of a normal sp^2 hybrid molecule is
 A. 109.5° B. 107.4° C. 104.5° D. 120° E. None of the above
48. Calculate the percent ionic character for the C-Cl bond in CCl_4 .
 $\Delta H_{f, C} = 83.1 \text{ kJ/mole}$, $\Delta H_{f, CCl_4} = 58.0 \text{ kJ/mole}$, Electronegativity, $\chi_C = 2.5$, Electronegativity, $\chi_{Cl} = 3.0$
 A. 6.2% B. 93.8% C. 27.7% D. 30.3% E. None of the above
49. Which of the following is the correct order of increasing first ionization energies for fluorine, sulphur, potassium, silicon and magnesium?
 A. Mg, K, Si, F, S B. F, Si, K, Mg C. K, Mg, Si, S, F D. K, Mg, S, Si, F
50. The incorrect statement about electron affinity is:
 A. the amount of energy released when an electron adds to an isolated gaseous neutral atom is called its electron affinity B. group 2 group 8 elements have zero or positive electron affinities C. electron affinities can have positive and negative values D. all are correct
51. Which of the following groups of atoms have been placed in order of increasing electronegativity?
 A. Na, Rb, K B. Mg, Ca, Sn C. B, Be, Li D. N, O, F
52. Which of the following statements about hydrides is incorrect?
 A. Ionic or salt-like hydrides e.g. LiH react with water to produce basic solutions B. All covalent hydrides dissolve in water to give acidic solutions C. CH_4 , HCl and HF are all gases at $25^\circ C$ D. $NaH(g) + H_2O(l) \rightarrow NaOH(aq) + H_2(g)$ is an oxidation-reduction reaction
53. The species O^{2-} , F^- , Ne , Na^+ and Mg^{2+} are isoelectronic. Which of the following statements is/are not correct?
 i. All five of the atoms or ions have the same number of protons in the nucleus. ii. It would require more energy to remove an electron from Na^+ than from Ne . iii. O^{2-} has a larger radius than any of the others. iv. Mg^{2+} is smaller than Ne . v. F^- has one electron more than Ne .
 A. (i), (iv) and (v) B. (ii), (iii) and (iv) C. (ii), (iii) and (v) D. (i) and (v) E. None of the above
54. The shape of a molecule or ion is the geometrical arrangement of

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- The central atom around the ligands B. the ligands around the central atom C. ions around the molecules D. ligands around the multiple bond E. None of the above.
55. The oxidation state of carbon in the ground state is
+4 B. +3 C. 0 D. +1 E. None of the above.
56. The force of repulsion in a bond is greatest between
A. bond pair and bond pair B. lone pair and bond pair C. lone pair and lone pair D. bond pair and the central atom E. None of the above
57. The overlapping of S-S atomic orbitals gives
A. spherical shape B. trigonal shape C. tetrahedral shape D. trigonal bipyramid E. None of the above
58. Which of the following elements has the greatest spin?
A. Hydrogen B. Helium C. Aluminum D. Nitrogen E. None of the above
59. Hybridization of the valence orbitals of carbon gives rise to
A. four degenerate orbitals B. three degenerate orbitals C. two degenerate orbitals D. five degenerate orbitals E. None of the above
60. Which of the following is not a period 3 element?
A. Silicon B. Titanium C. Aluminum D. Phosphorus E. None of the above
61. The four main types of electronic structure in the periodic table are
A. s, p, d & f blocks B. s, p, d & h blocks C. s, d, g, & h blocks E. None of the above
62. The elements with the electronic configuration, ns^2p^6 are
Chemically reactive B. chemically inert C. moderately reactive D. too reactive E. None of the above
63. Which of the following molecules has bond angle of 120° ?
A. SnCl_2 B. H_2O C. TeCl_4 D. NH_3 E. None of the above
64. Which one of the following comparisons about oxides is incorrect?
A. K_2O is more basic in character than CO_2 B. Both Al_2O_3 and SnO_2 are amphoteric
C. N_2O_5 is more acidic in character than H_2O D. Al_2O_3 is more basic in character than MgO
E. None of the above
65. Which is the most nonmetallic of the following?
A. Be B. Ga C. Br D. Mg E. None of the above
66. Which main group element(s) has (have) a different number of outermost electrons than its (their) group number?
A. Transition metals B. Halogen C. Chalcogen D. The noble gases E. None of the above
67. $^{140}_{54}\text{Ba}$ decays by β^- emission. Which nuclide would be the product of the decay process?
A. $^{140}_{55}\text{La}$ B. $^{140}_{56}\text{Ba}$ C. $^{140}_{53}\text{I}$ D. $^{140}_{52}\text{Te}$ E. None of the above
68. The three most common kinds of radioactive decay are
A. alpha, beta and positron emission B. beta, positron emission and gamma radiation
C. alpha, positron emission and gamma radiation D. gamma radiation, beta and alpha E. None of the above
69. Which of the following statements about nuclear reactions is incorrect?
A. the emission of a positron has the effect of converting a proton to a neutron
B. Electron capture has the effect of converting a proton to a neutron
C. A positron is a particle that has the same mass as an electron, but with an opposite charge
D. The emission of a positron has the effect of increasing the atomic number of the nucleus by one.
E. None of the above
70. Which of the following contains an error?
A. Alkali metal - Cs, Li B. The coinage metals are Au, Ag and Cu C. Transition elements, Sn and Pb can exhibit variable valencies D. Liquid halogens - I_2 , Br_2 E. None of the above

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